

Edges of Identity: A Reverse Turing Test

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Figure 1: AI-driven avatars discussing their agendas in the train's compartment.

ABSTRACT

In an era where artificial intelligence is fundamentally changing our perception of reality, 'Edges of Identity: The Reverse Turing Test' presents a fascinating reversal of the classic Turing Test. This immersive Virtual Reality (VR) installation challenges visitors to disguise themselves as the only human among advanced AI systems, raising profound questions about the nature of intelligence, identity and reality.

Index terms: Artificial intelligence, virtual reality, large language models, human-agent-interaction, Turing test.

1 PROJECT URL

<https://www.theater-an-der-ruhr.de/de/programm/5759-erweiterte-realitaten>

2 DESCRIPTION OF ARTWORK

'Edges of Identity: The Reverse Turing Test' is an interactive Virtual Reality installation that explores the boundaries between artificial and human intelligence, raising fundamental questions about human nature and reality in an increasingly digitalised world. At a time when AI systems and virtual humans are becoming

increasingly sophisticated [1, 2], our installation challenges participants to question their notions of authenticity, identity and human behaviour.

In our VR-Experience, human visitors find themselves in a virtual train compartment, embodying the persona of Genghis Khan as an avatar. The other passengers are famous personalities as well: Aristotle, Mozart, Leonardo da Vinci, and Cleopatra. However, they are driven by some of the most sophisticated AIs currently available: GPT4, Claude, Llama, and Gemini. The train conductor informs the group that one of them is a human and they have to find out who. Aristotle suggests that each one asks the person to their left a question that helps find out whether the one answering is a human or AI. Thus, the installation reverses the classic Turing test [3]: instead of a human trying to identify a machine, the AIs have to unmask the human among them.

This reversal creates a fascinating dynamic in which the human participant is forced to reflect on and adapt their behavior in order to 'pass' as an AI. This game of identity, imitation and projection raises profound questions:

- How do we define and recognise human behaviour in a world where AIs are becoming increasingly human-like?
- To what extent does our behaviour change in environments that are increasingly populated by artificial intelligences?

The installation invites visitors to explore the boundaries between human and machine, past and future, reality and simulation, while asking fundamental questions about the nature of our increasingly digitalized existence [4].

The experience was developed using the Unity Engine and runs on all headsets via OpenXR. At the start, each of the different characters is randomly assigned an LLM (for example, Aristotle is

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controlled by GPT-4o, Mozart by Claude Sonnet, etc.). Depending on who is currently up for questions/answers/voting, the corresponding LLM is prompted, and the answer from the respective character is then output via text-to-speech (Google and Azure) with real-time lip sync.

The prompt contains a description of the situation, the complete conversation so far, and the instruction. In the output, in addition to the spoken part, meta-information is provided in JSON—such as whether the current statement is an answer, a question to the next person, a vote, and if it is a vote, for whom, etc. The player's voice is transcribed into text by speech-to-text services, and then, along with the prompts, sent to the respective LLM.

The intro and outro are scripted; the LLMs only come into play when Aristotle poses the first question to Mozart.

3 ARTIST INFORMATION

Tore Knabe is a Berlin-based Virtual Reality developer and digital pioneer who consistently explores the frontiers of emerging technologies. His journey began with the dawn of mobile computing, where his innovative games and educational apps, including the globally successful *Bowmaster*, reached top positions in App Stores worldwide. Since 2013, Knabe has been at the forefront of Virtual Reality experimentation, creating numerous experiences that push the boundaries of human-computer interaction. His recent work focuses on the intersection of VR and artificial intelligence, demonstrating through compelling videos how AI-driven virtual characters might transform our digital interactions. Through these explorations, Knabe offers glimpses into the near future of human-machine communication and the evolving landscape of virtual experiences.

Jonathan Harth studied sociology, philosophy and psychology at the Free University of Berlin and the University of Vienna and has been working as a research assistant at the Chair of Sociology at Witten/Herdecke University since 2009. His research focuses on the sociology of digitalisation (extended reality and sociality under the conditions of machine intelligence), digital art production and the sociology of religion (Western Buddhism). In addition to his work at the Chair of Sociology, Harth works on the project 'Theatre of Extended Realities' (funding programme *Neue Wege*), where he is dedicated to the conception and use of digital technologies in virtual and augmented theatre plays.

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